Trigonometric functions

Lecture and worked examples

Objectives

Use parity and periodicity; differentiate sine and cosine; solve equations and inequalities; study a periodic function.

Circle and functions

On the unit circle, the point located by an angle x in radians has coordinates $(\cos x, \sin x)$. For every real number x,

$$\cos^2 x + \sin^2 x = 1, \quad -1 \le \cos x, \sin x \le 1.$$

Cosine is even: $\cos(-x) = \cos x$. Sine is odd: $\sin(-x) = -\sin x$. Both functions are 2π-periodic.

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1

Derivatives and variations

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x.$$

$$(\sin(ax+b))' = a\cos(ax+b), \quad (\cos(ax+b))' = -a\sin(ax+b).$$

On $[-\pi, \pi]$, cosine increases on $[-\pi, 0]$ from -1 to 1, then decreases on $[0, \pi]$ from 1 to -1. Sine decreases on $[-\pi, -\frac{\pi}{2}]$, increases on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, then decreases on $[\frac{\pi}{2}, \pi]$.

Derivatives and variations

Read the sign of sine and cosine on the circle, then apply the link between derivative and variations. Periodicity extends the curves; it does not remove the need to study an interval of sufficient length.

Equations and inequalities

If $|a| > 1$, $\cos x = a$ and $\sin x = a$ have no real solution. For a known angle θ,

$$\cos x = \cos\theta \iff x = \theta + 2k\pi \text{ or } x = -\theta + 2k\pi,$$

$$\sin x = \sin\theta \iff x = \theta + 2k\pi \text{ or } x = \pi - \theta + 2k\pi, \quad k \in \mathbb{Z}.$$

Then restrict the solutions to the required interval and eliminate duplicates.

Equations and inequalities

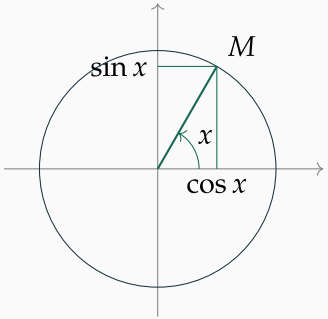
$\cos x = \frac{1}{2}$ gives $x = \pm\frac{\pi}{3}$; $\sin x = \frac{1}{2}$ gives $x = \frac{\pi}{6}$ or $\frac{5\pi}{6}$. $\cos x \leq \frac{1}{2}$ gives $[-\pi, -\frac{\pi}{3}] \cup [\frac{\pi}{3}, \pi]$; $\sin x \geq \frac{1}{2}$ gives $[\frac{\pi}{6}, \frac{5\pi}{6}]$.

To solve $\cos(2x) = \frac{1}{2}$ on $[-\pi, \pi]$, set $t = 2x \in [-2\pi, 2\pi]$, solve for t, then divide by 2. We obtain $x \in \{-\frac{5\pi}{6}, -\frac{\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}\}$.

For $f(x) = \sin x + \frac{1}{2}\sin(2x)$ on $[0, \pi]$,
$f' = \cos x + \cos(2x) = 2\cos^2 x + \cos x - 1 = (2\cos x - 1)(\cos x + 1)$. The function increases up to $\frac{\pi}{3}$ then decreases; maximum $\frac{3\sqrt{3}}{4}$.

For every real number x, $\sin(2x) = 2\sin x \cos x$ and $\cos(2x) = \cos^2 x - \sin^2 x = 2\cos^2 x - 1$. These identities are recalled here so that their use in the problems is not left implicit.

The evenness of cosine corresponds to a symmetry about the y-axis; the oddness of sine to a point symmetry about the origin. Periodicity extends each curve by translation of vector $(2\pi, 0)$. The coordinates of the point on the circle are $(\cos x, \sin x)$.



Height of a point on a wheel - 35 min.

A wheel turns at constant speed in a vertical plane. Its radius is 2 m and its centre is 3 m above the ground. A point on the wheel is at its lowest position at time $t = 0$. The wheel makes one revolution in 8 s. Its height is modelled by $h(t) = 3 - 2\cos\left(\frac{\pi t}{4}\right)$, where t is in seconds and $h(t)$ in metres.

Research activity (2)

1. Check that the model gives the stated initial height, a period of 8 s and the expected minimum and maximum heights.
2. Compute $h'(t)$, then study the variations of h on $[0; 8]$.
3. Determine all the times $t \in [0; 8]$ at which the height is 4 m.
4. Determine the total time, during this revolution, for which the point is at least 4 m above the ground.